MOHAMMAD R. AL-EIADEH

Graduate Research Assistant ∼ R & D

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SUMMARY

Mohammad Riyad Al-Eideh is a dedicated Ph.D. candidate in the Elmore family school of Electrical and Computer Engineering at Purdue University. He earned his B.Sc. and M.Sc. in Computer Science from Al-Balqa University in 2018 and from Yarmouk University in 2021. His **research** focuses on **Graph ML, Graph Computations, Matrix Perturbation Theory, Dynamic Networks, and Cybersecurity (Interdependent Systems modeled as Attack Graphs)**. He has experience on optimization problems such as Graph Embedding. He is interested in learning on graphs, Java, multi-threading, functional programming, and data structures, and is passionate about solving complex challenges in computer science. Contact him at maleiade@purdue.edu.

SKILLS

Languages: Java, C++, C#, Python.

Technologies: EJML (Matrix Library), Tribuo (ML library), DFLib (DataFrame library), JGraphT library, XChart library, Open CSV API, JDBC, Collection Framework, Stream & Parallel Stream, Executor Service, Maven.

Programming Paradigms: OOP, Packaging, Design Pattern, Generics, Functional | Declarative Programming.

Research Interests: Graph ML, Cyber Security, Matrix Computation, Network Science, Complex Networks & Time Complexity Analysis.

EDUCATION

8/2023 - present **Purdue University, Indiana, USA** **School**
Ph.D. in Electrical and Computer Engineering, 90 credit hours
Thesis Fields: Cybersecurity, Graph Algorithms, Attack Graphs, Resource Allocation, Machine Learning

9/2019 - 7/2022 **Yarmouk University, Irbid, Jordan** **School**
Masters' in Computer Science, 32 Credit Hours
Thesis Fields: Feature Selection. Evolutionary Algorithm. Machine Learning

9/2015 - 7/2018 **Al-Huson University College | Al-Balqa Applied University, Irbid, Jordan** **School**
Bachelors' in Computer Science, 132 Credit Hours
Project Fields: Android, Arduino, Smart Home

EXPERIENCE

8/2023 – Present **Graduate Researcher & Software Developer** **Advisor: Dr. Mustafa Abdallah**

- **Securing Interdependent Systems**: Developing graph-theoretic and graph-learning-based models for protecting interdependent systems represented as attack graphs.

1/2026 – Present **Graduate Teaching Assistant** **Courses: CIT 44000, CIT 40600 — Purdue University**

- Assisting in **Communication Network Design** & **Advanced Network Security**; grade homework and laboratory assignments, Provide academic support and hold office hours.

8/2025 – 12/2025 **Graduate Teaching Assistant** **Course: CNIT 25501 — Purdue University**

- Assisted in **Java programming**, mentoring students in **OOP, basic GUI development, and problem solving**; evaluated coursework and provided technical support during lab sessions.

1/2025 – 5/2025 **Graduate Teaching Assistant** **Purdue University**

- Mentored students in programming laboratories, assisting with **data structures, algorithms, and debugging**, and providing technical guidance during lab sessions.

8/2024 – 12/2024 **Graduate Teaching Assistant** **Course: ENGR133 — Purdue University**

- Supported instruction in **Python and MATLAB** for first-year engineering students, helping develop computational thinking and engineering problem-solving skills.

8/2022 – 8/2023 **Researcher & Software Developer** **Advisor: Prof. Mohammad Batah**

- Developed hybrid **evolutionary optimization algorithms** for NP-hard combinatorial problems, including **Maximum Clique (DIMACS)**, **Feature Selection** & **0–1 Knapsack**.

AWARDS & HONORS

7/2023 **Purdue School of Engineering and Technology, IUPUI, IN, USA**
Graduate Research Assistant-ship, PhD in ECE at E & T Department

6/2018 **Al-Huson University College, Al-Huson, Jordan**
Academic Excellence Award - Ranked 1st in my class for academic performance

PHD COURSES

ECE 62900 – INTRO NEURAL NETWORKS
ECE 66200 – PATTN RECOG / DECIS PROC
MA 51100 – LINEAR ALGEBRA APPL
ECE 58000 – OPT METH FOR SYS & CON
ECE 59500 – INTRODUCTION TO GAME THEORY
MA 57500 – GRAPH THEORY

PUBLICATIONS

Journal Article — **Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah. A Robustness Analysis of Attack-Graph-Driven Security Resource Allocation Schemes under Adversarial Attacks. Under Processing at ACM Transactions on Privacy and Security (TOPS).** ***Under Review***

Journal Article — **Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah. Auto-Sec: Meta-Learning for Automated Efficient Security Resource Allocation on Attack Graphs. IEEE Transactions on Dependable and Secure Computing, 2025.** GitHub

Conference Paper — **Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah. SAG-ERC: Securing Attack-Graph-based Systems through Node Embedding, Ranking, and Clustering. IEEE Conference on Secure and Trustworthy CyberInfrastructure for IoT and Microelectronics (SaTC), Ohio, Dayton, USA, 2025.** GitHub

Journal Article — **Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah. CBDRA-IS: Centrality-Based Defense Resource Allocation for Securing Interdependent Systems. ACM Trans. Priv. Secur., 2025.** GitHub

Journal Article — **Al-Batah, M. S., Mohammad Ryiad Al-Eiadeh, & Alnsour, Y. A Novel Approach for Enhancing Plant Leaf Classification With the Binary Cuckoo Search Algorithm. Applied Computational Intelligence and Soft Computing, 2025.** GitHub

Journal Article — **Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah. PR-DRA: PageRank-based Defense Resource Allocation Methods for Securing Interdependent Systems Modeled by Attack Graphs. International Journal of Information Security, 2025.** GitHub

Conference Paper — **Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah. AARA-PR: Asset-Aware PageRank-Based Security Resource Allocation Method for Attack Graphs. IEEE Interdisciplinary Conference on Electrics and Computer (INTCEC), IEEE, Chicago, IL, USA, 2024.** GitHub

Journal Article — **Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah. Genigraph: A Genetic-Based Novel Security Defense Resource Allocation Method for Interdependent Systems Modeled by Attack Graphs. Computers & Security, 2024.** GitHub

Journal Article — **Al-Batah, M. S., & Mohammad Ryiad Al-Eiadeh. An Improved Binary Crow-JAYA Optimization System With Various Evolution Operators Such as Mutation for Finding the Max Clique in Dense Graphs. International Journal of Computing Science and Mathematics, 2024.** GitHub

Journal Article — **Mohammad Ryiad Al-Eiadeh, Raneem Qaddoura, and Mustafa Abdallah. Investigating the Performance of a Novel Modified Binary Black Hole Optimization Algorithm for Enhancing Feature Selection. Applied Sciences, 2024.** GitHub

Journal Article — **Al-Batah, M. S., & Mohammad Ryiad Al-Eiadeh. An Improved Discrete Jaya Optimisation Algorithm With Mutation Operator and Opposition-Based Learning to Solve the 0-1 Knapsack Problem. International Journal of Mathematics in Operational Research, 2023.** GitHub

Journal Article — **Mohammad Ryiad Al-Eiadeh. Automatic Lung Field Segmentation Using Robust Deep Learning Criteria. International Journal of Hybrid Information Technologies, 2021.**

– POSTER PRESENTATIONS

2026 **CERIAS Annual Security Symposium, Purdue University** — **Poster Presentation**
Mohammad Al-Eiadeh and Mustafa Abdallah. *CBDRA-IS: Centrality-Based Defense Resource Allocation for Securing Interdependent Systems.* Poster

2025 **CERIAS Annual Security Symposium, Purdue University** — **Poster Presentation**
Mohammad Al-Eiadeh and Mustafa Abdallah. *PR-DRA: PageRank-based Defense Resource Allocation Methods for Securing Interdependent Systems Modeled by Attack Graphs.* Poster

PROFESSIONAL SERVICE

2026 **IEEE** — **Journal Peer Reviewer**
Reviewed **1 manuscript** for *IEEE Transactions on Information Forensics and Security (TIFS).*

2026 **Springer Nature** — **Journal Peer Reviewer**
Reviewed **1 manuscript** for *Journal of King Saud University Computer and Information Sciences.* Certificate

2026 **Springer Nature** — **Journal Peer Reviewer**
Reviewed **1 manuscript** for *International Journal of Machine Learning and Cybernetics.* Certificate

2025 **Springer Nature** — **Journal Peer Reviewer**
Reviewed **1 manuscript** for *Cluster Computing.* Certificate

2025 **Springer Nature** — **Journal Peer Reviewer**
Reviewed **1 manuscript** for *Applied Network Science.* Certificate

REFERENCE

Scholar **Purdue University, Indianapolis & West Lafayette, IN, USA** — **Supervisor**
Prof. Mustafa Abdallah
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Instructor **Purdue University, Indianapolis & West Lafayette, IN, USA** — **Assistant & Cordinator**
Sherrie Tucker
Email: tucker76@purdue.edu

PROJECTS

Software Project **GraphEmbedding4j: In-progress Java library for graph / node embedding research, with modular components for random walks, context-window extraction, negative sampling, skip-gram learning, gradient descent optimization, and sigmoid-based loss modeling.** **GitHub**

LANGUAGES

Arabic - **Native**, **English** - **Fluent**